

Unit 04 Practice Problems: Discrete Random Variables

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Required Practice Problems

Complete these required practice problems throughout the unit as we cover the corresponding topics in class. These problems reinforce the concepts discussed in the lecture notes, handouts, and course resources. Before attempting the homework assignments and preparing for exams, make sure you have completed and reviewed all practice problems from the unit. The **solutions** are posted on Canvas ([Unit-PP-Solution.pdf](#) > [Practice Problem-Solution](#) > [STAT 312](#) > [Files](#)).

4.1

A basket contains 3 green and 2 yellow balls. One ball will be selected at random and then not replaced. Then a second ball will be randomly selected from the basket.

Let

X = number of green balls observed during the experiment.

(a) Draw a tree diagram with probabilities written on the branches. At the end of each branch, identify each outcome of the Sample Space and its probability.

(b) Write the PMF (probability mass function) of X .

4.2

Refer to Practice Problem 3.17.

There are 21 balls in a basket: 10 are red, 6 are green, 3 are yellow, and 2 are orange. Four balls will be drawn from the basket all at once at random without replacement.

Let

X = number of orange balls observed.

Find the PMF (probability mass function) of X .

4.3

Refer to Practice Problem (Unit 03) 3.12.

An unfair coin has

$$P(H) = 0.68.$$

The coin will be flipped three times. Let

X = number of heads.

Find the PMF (probability mass function) of X .

4.4

A discrete random variable X has the following PMF:

x	$p(x)$
1	0.60
3	0.20
7	0.10
11	0.10

(a) Find

$$P(X > 6).$$

(b) What is the probability that two independent observations of X will both equal 1?

(c) Find the population mean μ_X , also known as

$$E(X),$$

the expected value of X .

(d) Find the population variance of X using both the “regular” formula and the “convenient” formula:

$$\sigma_X^2 = \sum (x - \mu_X)^2 p(x) = E(X^2) - \mu_X^2.$$

Make a table and show your work. You should obtain the same answer using both formulas. Finally, find the population standard deviation σ_X .

4.5

Refer to Practice Problem 3.13.

If

X = the outcome of one roll of the unfair die,

find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X.$$

4.6

A large group of people need to have their blood tested to determine whether or not they have Disease X. Instead of testing everyone individually, people are placed into groups of 10.

The blood samples of the 10 people will be pooled and analyzed together. If the test is negative, we conclude that 0 of the 10 people have the disease and no more testing is required.

If the test is positive, we conclude that at least one person in the group has the disease, and all 10 people must then be individually tested, for a total of 11 tests.

Let

X = total number of tests needed for a group.

What are the possible values of X ?

Find $E(X)$ if the probability that any person has the disease is p .

Then find $E(X)$ if

$$p = 0.02.$$

4.7

Refer to Practice Problems 3.13 and 4.5.

(a) Suppose the integers

$$\{1, 2, 3, 4, 5, 6\}$$

were removed from the six faces of the unfair die and replaced with

$$\{11, 12, 13, 14, 15, 16\}.$$

Find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X,$$

and compare these values to what you found earlier.

(b) Suppose the integers

$$\{1, 2, 3, 4, 5, 6\}$$

were removed from the six faces of the unfair die and replaced with

$$\{10, 20, 30, 40, 50, 60\}.$$

Find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X,$$

and compare these values to what you found earlier.

4.8

Refer to Practice Problem 4.3.

An unfair coin has

$$P(H) = 0.68.$$

The coin will be flipped three times. Let

X = number of heads.

Is X a Binomial random variable? Explain why or why not.

If it is, identify the values of the parameters n and p , then find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X.$$

4.9

Julian is shooting arrows at a target in the backyard. The probability that he hits the bull's-eye is 0.35 on every attempt, and all of his attempts are independent of each other.

Let

X = number of successful bull's-eyes out of $n = 30$ attempts.

(a) Find

$$P(X = 9).$$

(b) Find

$$P(X = 12).$$

(c) Find

$$P(X \geq 16).$$

Express your answer using summation notation, then use a calculator or computer to find the answer.

(d) Find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X.$$

4.10

An eight-sided die has eight faces with the integers

$$\{1, 2, 3, 4, 5, 6, 7, 8\}$$

equally likely to appear when the die is rolled.

- (a) If the die is rolled five times, what is the probability that an “8” appears exactly two times?
- (b) If the die is rolled six times, what is the probability that an even number appears more than four times?
- (c) If the die is rolled seven times, what is the probability that a “6”, “7”, or “8” appears on exactly four of the seven rolls?

4.11

Jeremiah and Crystal are basketball players who like to compete against each other. They are about to attempt 10 free-throw shots each.

Jeremiah’s probability of a successful free-throw attempt is always 0.74.

Crystal’s probability of a successful free-throw attempt is always 0.84.

Jeremiah and Crystal are arguing about which event is more likely:

- Jeremiah will have 8 or more successful free throws out of 10 attempts.
- Crystal will have 9 or more successful free throws out of 10 attempts.

Who is correct?

4.12

Refer to Practice Problem 4.9.

Every day for a year, Julian shoots arrows at the target and writes down how many attempts were needed to hit the bull’s-eye for the first time. At the end of the year, Julian has a data set with 365 observations.

- (a) Approximately what proportion of the 365 observations will be zeros? Ones? Twos?
- (b) If Julian calculates $\bar{X}$, s^2 , and s for the data set, they will be approximately equal to what values?

4.13

Jodie is very talented at throwing darts. The probability that she successfully hits the bull’s-eye equals 0.79. Assume that this probability of success never changes and that all of her throws are independent of each other.

Jodie will begin throwing darts at the bull’s-eye.

- (a) What is the probability that Jodie’s first success will occur on her third attempt?
- (b) What is the probability that Jodie’s first success will not occur on her first or second attempt?

4.14

Refer to Practice Problem 4.11.

Now Jeremiah and Crystal will each attempt free throws until they observe their first success.

They are arguing about which event is more likely:

- Jeremiah's first success will occur on his first or second attempt.
- Crystal's first success will occur on her first attempt.

Who is correct?

4.15

Optional Challenge Problem

An unusual vending machine works like this: if you insert \$1 into the machine, it will give you a playing card at random from a standard 52-card deck.

Cards are dispensed "with replacement," so, for example, it is possible that the vending machine could give you the Ace of Spades two times in a row.

Suppose you are going to stand at this vending machine and insert \$1 bills until you have a complete deck of all 52 different cards.

Let

X = number of \$1 bills needed to acquire a complete deck.

Easy question: What are the possible values of X ?

Difficult question: What is the expected value of X ?

4.16

Refer to Practice Problem 4.10.

(a) What is the probability that the second occurrence of a "7" will occur on the 5th roll of the eight-sided die?

(b) What is the probability that the fourth occurrence of an even number will occur on the 8th roll of the eight-sided die?

(c) Suppose every time a 3, 4, or 5 is rolled, you win a prize. What is the probability that you will be awarded your third prize on the 7th roll of the eight-sided die?

4.17

Jeff is friends with Jodie (refer to Practice Problem 4.13), and he hits the bull's-eye with probability 0.40. Assume that this probability of success never changes and that all of his throws are independent of each other.

Jeff will begin throwing darts at the bull's-eye.

(a) What is the probability that Jeff's fourth success will occur on his tenth attempt?

(b) What is the probability that Jeff will need to throw 4 or more darts in order to observe his second bull's-eye?

4.18

Refer to Practice Problem 3.12.

An unfair coin has

$$P(H) = 0.68.$$

Suppose the coin will be flipped until heads is observed for the 100th time.

Let

X = number of flips needed.

Find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X.$$

4.19

Let

X = number of parking tickets a campus parking officer writes on any particular day.

Assume that X is a Poisson random variable with mean

$$\lambda = 9.3$$

tickets per day.

- (a) What is the probability that the parking officer will write exactly 7 tickets tomorrow?
- (b) Every day for a year, the parking officer writes down how many tickets he wrote that day. At the end of the year, he has a data set with 365 observations. If he calculates $\bar{X}$, s^2 , and s for the data set, they will be approximately equal to what values?

4.20

Mrs. Hopkins enjoys watching hummingbirds approach her hummingbird feeder on a summer afternoon.

Let

X = number of hummingbirds that approach the feeder per minute.

Assume that X is a Poisson random variable with mean

$$\lambda = 4.7$$

hummingbirds per minute.

- (a) What is the probability that exactly 3 hummingbirds will approach her feeder within the next minute?
- (b) What is the probability that more than 2 hummingbirds will approach her feeder within the next minute?

4.21

An island in the South Pacific experiences frequent major earthquakes.

Let

X = number of major earthquakes per 10-year period.

Assume

$$X \sim \text{Poisson}(\lambda = 3.16).$$

(a) What is the probability that there will be at least one earthquake within the next year?

Hint: Notice that this question is asking about a one-year period, not a 10-year period.

(b) What is the probability that there will be fewer than five earthquakes within the next 15 years?

4.22

Refer to Practice Problem 4.2.

What type of discrete random variable is

X = number of orange balls observed?

Identify the values of the parameter(s), then find

$$\mu_X, \quad \sigma_X^2, \quad \sigma_X.$$

4.23

A large classroom has 68 students:

- 19 are biology majors,
- 24 are chemistry majors,
- 25 are math majors.

(a) If ten students are randomly selected all at once and without replacement, what is the probability that exactly 4 are math majors?

(b) If three students are randomly selected all at once and without replacement, what is the probability that they all have the same major?

4.24

A basket contains 5 orange, 6 yellow, and 7 white balls.

Identify each of the following random variables as **Binomial**, **Geometric**, **Negative Binomial**, or **Hypergeometric**. Also identify the values of the parameter(s).

(a) Draw 5 balls all at once at random without replacement.

A = number of white balls.

(b) Draw 5 balls one at a time at random with replacement.

B = number of white balls.

(c) Draw balls from the basket one at a time at random with replacement until you see the first yellow ball.

C = number of draws needed.

(d) Draw balls from the basket one at a time at random with replacement until you see the third yellow ball.

D = number of draws needed.

(e) Draw balls from the basket one at a time at random with replacement until you see a ball that is not orange.

E = number of draws needed.

(f) Draw 10 balls all at once at random without replacement.

F = number of balls that are not orange.

4.25

Optional Challenge Problem

Refer to Practice Problem 4.4.

- (a) Write the moment generating function of X .
- (b) Use $M(t)$ to find the first and second moments of X and confirm that you obtain the same answers as in Practice Problem 4.4.
- (c) What is the probability that seven independent observations of X will add up to 33?

Advice: Use Mathematica.

4.26

Solve the following exercises from the “Chapter 3 Exercises” section of the textbook (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger).

- 3.1.1, 3.1.3, 3.1.5, 3.1.6, 3.2.1, 3.2.3, and 3.2.5.

[To find these exercises in the digital copy of the textbook (Textbooks > STAT 312 > Files > Canvas), go to the xvi Contents page and click the “EXERCISES” tab located at the lower-right corner of the page. This will open the complete list of chapter exercises. The **solutions** are posted on Canvas (textbook-solution.pdf > Practice Problem-Solution > STAT 312 > Files).]

Supplemental Practice Problems

The problems below provide additional practice opportunities for students who would like more practice with the concepts covered in this unit.

Chapter 2 (*Applied Statistics and Probability for Engineers*, 7th edition, by Montgomery and Runger)

- 3.1.9, 3.1.11, 3.1.13, 3.2.7, 3.3.1, 3.3.3, 3.3.5, 3.4, 3.5.1 (a-d), 3.5.3, 3.5.7, and 3.5.13